```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import numpy as np
```

```
import pandas as pd

#Data processing

path = '/content/drive/MyDrive/Data Analysis Project/Project 1/'
patients=pd.read_csv(path + 'patients.csv')
diagnoses = pd.read_csv(path + 'diagnoses.csv')
outcomes=pd.read_csv(path + 'outcomes.csv')
labs=pd.read_csv(path + 'labs.csv')
```

```
df = (patients
      .merge(diagnoses, on='DiagnosisID', how='left')
      .merge(outcomes, on='OutcomeID', how='left'))

df.head()
```

	PatientID	Name	Age	Gender	DiagnosisID	AdmissionDate	DischargeDate	OutcomeID	TreatmentCost	DiagnosisName	Ou
0	1	Edward Montes	68	M	2	2025-03-22	2025-04-04	1	4443	Diabetes	
1	2	Jessica Zimmerman	81	M	2	2025-03-31	2025-04-02	2	4265	Diabetes	C
2	3	Gina Alexander	58	M	6	2025-01-08	2025-03-13	1	4188	COPD	
3	4	Lori Green	44	F	3	2025-04-03	2025-04-05	1	9783	Heart Disease	
4	5	Jack Logan	72	F	9	2025-06-03	2025-06-04	1	5005	Kidney Disease	

Next steps: Generate code with df | New interactive sheet

```
df['AdmissionDate']=pd.to_datetime(df['AdmissionDate'])
df['DischargeDate']=pd. to_datetime(df['DischargeDate'])
df['Lengthofstay']=(df['DischargeDate']-df['AdmissionDate']).dt.days
```

```
df['OutcomeEncoded']=df['OutcomeName'].map({'Recovered':0,'Complicated':1,'Deceased':1})
```

```
df['HighRisk']=np.where((df['Age']>65)|(df['OutcomeName'].isin(['Complicated','Deceased'])),1,0)
```

```
abnormal_conditions = {
    'Blood Sugar': lambda x: x > 120,
    'Cholesterol': lambda x: x > 200,
    'Hemoglobin':  lambda x: x < 13
}

def count_abnormal_labs(patient_id):
    patient_labs = labs[labs['PatientID'] == patient_id]
    count = 0
    for test_name, condition in abnormal_conditions.items():
        test_results = patient_labs[patient_labs['TestName'] == test_name]
        count += test_results['Result'].apply(condition).sum()
    return count                        # aligned with "for", so it runs after the loop

df['AbnormalLabcount'] = df['PatientID'].apply(count_abnormal_labs)
```

```
#Model Training

features = df[['Age', 'Lengthofstay','TreatmentCost']]
target = df['OutcomeEncoded']
```

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(features, target, test_size = 0.3, random_state=42)
```

```
from sklearn.linear_model import LogisticRegression
model = LogisticRegression(max_iter = 500)
model.fit(X_train, y_train)
```

```
LogisticRegression
LogisticRegression(max_iter=500)
```

```
from sklearn.metrics import classification_report, confusion_matrix
y_pref = model. predict(X_test)
print("Classification Report: ")
print(classification_report(y_test, y_pref))
```

```
Classification Report:
              precision    recall  f1-score   support

           0       0.00      0.00      0.00        17
           1       0.72      1.00      0.83        43

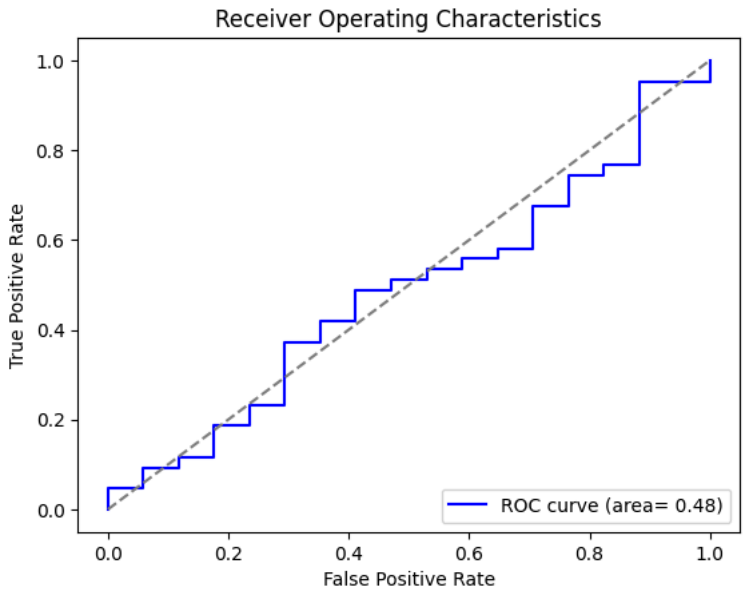
    accuracy                           0.72        60
   macro avg       0.36      0.50      0.42        60
weighted avg       0.51      0.72      0.60        60

/usr/local/lib/python3.13/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-de
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.13/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-de
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.13/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-de
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

```
import matplotlib.pyplot as plt
from sklearn.metrics import roc_curve, auc

y_prod = model.predict_proba(X_test)[:,1]
fpr,tpr, thresholds = roc_curve(y_test, y_prod)
roc_auc =auc(fpr,tpr)

plt.figure()
plt.plot(fpr, tpr,color = 'Blue', label = f'ROC curve (area= {roc_auc:.2f})')
plt.plot([0,1],[0,1], color= 'gray', linestyle = '--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Receiver Operating Characteristics')
plt.legend(loc='lower right')
plt.show()
```



Double-click (or enter) to edit

```
import joblib
joblib.dump(model, '/content/Risk_model1.pkl')
```

```
['/content/Risk_model1.pkl']
```

```
joblib.dump(model, path+'Risk_model1.pkl')
```

```
['/content/drive/MyDrive/Data Analysis Project/Project 1/Risk_model1.pkl']
```

```
!ls -la /content/Risk_model1.pkl
```

```
-rw-r--r-- 1 root root 1183 Oct 31 19:05 /content/Risk_model1.pkl
```

```
pip install streamlit
```

```
Collecting streamlit
  Downloading streamlit-1.66.1-py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: altair!=5.4.0,!=5.4.1,<7,>=5.0.0 in /usr/local/lib/python3.13/dist-packages (from streamlit)
Requirement already satisfied: click<9,>=7.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (8.5.0)
Requirement already satisfied: numpy<3,>=1.25.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.1.3)
Requirement already satisfied: packaging>=20 in /usr/local/lib/python3.13/dist-packages (from streamlit) (26.3)
Requirement already satisfied: pandas<4,>=1.5.3 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.2.3)
Requirement already satisfied: pillow<14,>=9.2.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (11.3.0)
Collecting pydeck<1,>=0.8.0b4 (from streamlit)
  Downloading pydeck-0.9.3-py2.py3-none-any.whl.metadata (4.2 kB)
Requirement already satisfied: protobuf<8,>=5.26.1 in /usr/local/lib/python3.13/dist-packages (from streamlit) (6.33.6)
Requirement already satisfied: pyarrow!=25.0.0,<27,>=10.0.1 in /usr/local/lib/python3.13/dist-packages (from streamlit) (23.
Requirement already satisfied: requests<3,>=2.27 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.32.4)
Requirement already satisfied: typing-extensions<5,>=4.10.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (4.1
Requirement already satisfied: starlette<2,>=0.46.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (1.10.0)
Requirement already satisfied: uvicorn<1,>=0.30.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (0.54.0)
Requirement already satisfied: anyio<5,>=4.0.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (4.15.1)
Requirement already satisfied: python-multipart<1,>=0.0.10 in /usr/local/lib/python3.13/dist-packages (from streamlit) (0.0.
Requirement already satisfied: websockets<18,>=12.0.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (15.0.1)
Requirement already satisfied: itsdangerous<3,>=2.1.2 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.2.0)
Requirement already satisfied: watchdog<7,>=2.1.5 in /usr/local/lib/python3.13/dist-packages (from streamlit) (6.0.0)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.13/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=5.0.0->str
Requirement already satisfied: jsonschema>=3.0 in /usr/local/lib/python3.13/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=5
Requirement already satisfied: narwhals>=1.14.2 in /usr/local/lib/python3.13/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.13/dist-packages (from anyio<5,>=4.0.0->streamlit) (3.20)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.13/dist-packages (from pandas<4,>=1.5.3->str
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.13/dist-packages (from pandas<4,>=1.5.3->streamlit) (2
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.13/dist-packages (from pandas<4,>=1.5.3->streamlit)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.27->
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.27->stream
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.27->stream
Requirement already satisfied: h11>=0.8 in /usr/local/lib/python3.13/dist-packages (from uvicorn<1,>=0.30.0->streamlit) (0.1
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.13/dist-packages (from jinja2->altair!=5.4.0,!=5.4.
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.13/dist-packages (from jsonschema>=3.0->altair!=5.4.0
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.13/dist-packages (from jsonsch
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.13/dist-packages (from jsonschema>=3.0->altair!
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.13/dist-packages (from jsonschema>=3.0->altair!=5.4
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.8.2->pandas<4,>=
Downloading streamlit-1.66.1-py3-none-any.whl (10.4 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 10.4/10.4 MB 54.0 MB/s  0:00:00
Downloading pydeck-0.9.3-py2.py3-none-any.whl (11.4 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 11.4/11.4 MB 73.7 MB/s  0:00:00
Installing collected packages: pydeck, streamlit
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2/2 [streamlit]
Successfully installed pydeck-0.9.3 streamlit-1.66.1
```

```
%%writefile app.py
import streamlit as st
import pandas as pd
import joblib

model = joblib.load('Risk_model1.pkl')

st.title("Healthcare Risk Stratification App")

age = st.number_input("Age", min_value=0)
length_of_stay = st.number_input("Length of stay (days)", min_value=0)
treatment_cost = st.number_input("Treatment cost", min_value=0.0)

if st.button("Predict"):
    input_data = pd.DataFrame(
        [[age, length_of_stay, treatment_cost]],
        columns=['Age', 'Lengthofstay', 'TreatmentCost']
    )
    prediction = model.predict(input_data)[0]
    probability = model.predict_proba(input_data)[0][1]

    st.write(f"Risk Prediction: {'High Risk' if prediction == 1 else 'Low Risk'}")
    st.write(f"Risk Probability: {round(probability, 2)}")
```

```
Overwriting app.py
```

```
!streamlit run app.py &>/content/logs.txt &
```

```
!cat /content/logs.txt
```

```
Collecting usage statistics. To deactivate, set browser.gatherUsageStats to false.

2026-10-31 19:14:21.700 Uvicorn server started on :::8502
```

```
!curl ipv4.icanhazip.com
```

```
34.106.167.83
```

```
!cat /content/logs.txt
!ps aux | grep streamlit
```

```
Collecting usage statistics. To deactivate, set browser.gatherUsageStats to false.

2026-10-31 19:14:21.700 Uvicorn server started on :::8502

  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8502
  Network URL: http://172.28.0.12:8502
  External URL: http://34.106.167.83:8502

root        4057  1.4  1.7 1600908 236176 ?      Sl   19:06   0:07 /usr/bin/python3 /usr/local/bin/streamlit run app.py
root        6072 30.2  0.5 451980 72412 ?        Sl   19:14   0:02 /usr/bin/python3 /usr/local/bin/streamlit run app.py
root        6118  0.0  0.0   7344  3552 ?        S    19:14   0:00 /bin/bash -c ps aux | grep streamlit
root        6120  0.0  0.0   6548  2328 ?        S    19:14   0:00 grep streamlit
```

```
!ls /content    # check that app.py and Risk_model1.pkl are listed
!nohup streamlit run app.py --server.port 8501 > /content/logs.txt 2>&1 &
import time; time.sleep(10)
!cat /content/logs.txt
!curl -s -o /dev/null -w "%{http_code}" http://localhost:8501
```

```
app.py  cf.txt  cloudflared  drive  logs.txt  Risk_model1.pkl  sample_data

Collecting usage statistics. To deactivate, set browser.gatherUsageStats to false.

2026-10-31 19:14:33.303 Port 8501 is not available
200
```

```
!nohup ./cloudflared tunnel --url http://localhost:8501 > /content/cf.txt 2>&1 &
import time; time.sleep(8)
!grep -o 'https://[a-z0-9-]*\.trycloudflare\.com' /content/cf.txt
```

```
https://employed-charge-does-mainland.trycloudflare.com
```